

## Abstract Algebra:

Group: A set  $G$  with operation  $*$  is a group if it satisfies:

1. Closure:  $a * b \in G$
2. Associativity:  $(a * b) * c = a * (b * c)$
3. Identity:  $a * e = a$
4. Inverse:  $a * a^{-1} = e$

## Modular Arithmetic:

Formula:  $a \equiv b \pmod{n}$

Example:  $17 \equiv 5 \pmod{12}$

Since  $17 - 5 = 12$ , divisible by 12

Number Theory: Studies properties of integers, especially prime numbers, divisibility and congruences.

$\gcd(a, b)$

$$\gcd(18, 24) = 6$$

RSA Algorithm: RSA is an asymmetric encryption algorithm based on prime factorization.

Modulus:  $n = p \times q$

Encryption:  $c = m^e \bmod n$

Decryption:  $m = c^d \bmod n$

Example: If  $m = 7, e = 5, n = 33$

$$c = 7^5 \bmod 33 = 10$$

ECC (Elliptic Curve Cryptography)

ECC uses points on an elliptic curve over a finite field.

$$\text{Eqn: } y^2 = x^3 + ax + b$$

Example:  $a = 2, b = 3$

$$y^2 = x^3 + 2x + 3$$



Hash Algorithm: It maps data to a fixed sized output.

$$h = H(m)$$

where,  $h$  = hash value

$m$  = message

Using SHA - 256

Hash algorithm used in password protection and data integrity.

IPsec (Internet Protocol security): IPsec

is a network-layer security protocol used to secure IP communications by authentication, encryption and integrity protection of IP packets.

Modes:

Transport mode: Protects payload only.

Tunnel mode: Protects entire IP packet.

## Diffie-Hellman key exchange mechanism

Public parameters: prime  $p$ , base  $g$

Alice computes:

$$A = g^a \bmod p$$

Bob computes:

$$B = g^b \bmod p$$

shared key:  $K = B^a \bmod p = A^b \bmod p$

Example: If  $p = 23$ ,  $g = 5$  Alice chooses

$a = 6$ , Bob chooses  $b = 15$ .

both obtain the same shared key

MAC (Message Authentication Code)

A MAC ensures message integrity and authenticity using a shared secret key.

$$MAC = F(k, m)$$



## Pseudo Random Number Generation

$$x_{i+1} = ax_i + c$$

$x$  is the sequence of Pseudo-Random Numbers

$m, (m > 0)$  the modulus

$a (0, m)$  the multiplier

$c (0, m)$  the increment

$x_0 [0, m)$  - Initial value